Department of Civil Engineering, National Cheng Kung University

Behavior and Design of Reinforced Concrete

Spring 2026 **Chung-Chan Hung**

Use OpenSEES, Response 2000, NewRC-Mocur2020, ETABS, SAP2000, ATENA, ABAQUS, or other cross-sectional analysis software (e.g. XTRACT) to analyze the problem below. Detailed design parameters shall be reasonably assumed and clearly presented in oral reports on the dates assigned.

Homework #1: Monotonic Axial Behavior of Columns (Due on 3/16 and 3/30)

1) Select a rectangular section.
2) Discuss the influences of two of the following parameters on the monotonic axial response of the section: reinforcing ratio, concrete strength, steel grade, and eccentricity.
3) Justify the accuracy of your numerical results.

3/16: Give an oral presentation (5-10 mins): what software will be used and why is it chosen?
3/30: Give oral presentation (10-15 mins) & Send slides to the instructor and teaching assistants.

Homework #2: Moment-Curvature Behavior (Due on 4/20)

1) Select a rectangular section.
2) Discuss the influences of two of the following parameters on the moment-curvature response of the section: reinforcing ratio, axial load ratio, concrete strength, and steel grade.
3) Justify the accuracy of your numerical results.

4/20: Give oral presentation (10-15 mins) & Send slides to the instructor and teaching assistants.

Homework #3: Monotonic Lateral (Pushover) Behavior of Columns (Due on 5/11)

HW#3: An RC column under gradually increased lateral displacement

1) Plot the load-displacement response.
2) Discuss the influences of TWO different concrete material models on the behavior of columns
3) Discuss the influences of TWO different steel material models on the behavior of columns
4) Give an example to discuss the Mander's confinement effect on the simulation results.
5) Justify the accuracy of your simulation results.

5/11: Give oral presentation (10-15 mins) & Send slides to the instructor and teaching assistants.

Term Project: Cyclic Lateral Behavior of Columns (Due on 6/1)

Term Project: An RC column under lateral displacement reversals

1) Plot the load-displacement response.
2) Discuss the influences of the materials or geometrical nonlinearity on the behavior of the column
3) Give an example to discuss the Mander's confinement effect on the simulation results.
4) Justify the accuracy of your simulation results.

6/1: Give oral presentation (10-15 mins) & Send slides to the instructor and teaching assistants.

Note: If you have any good ideas to make the homework practice and term project more interesting, feel free to change the problems.